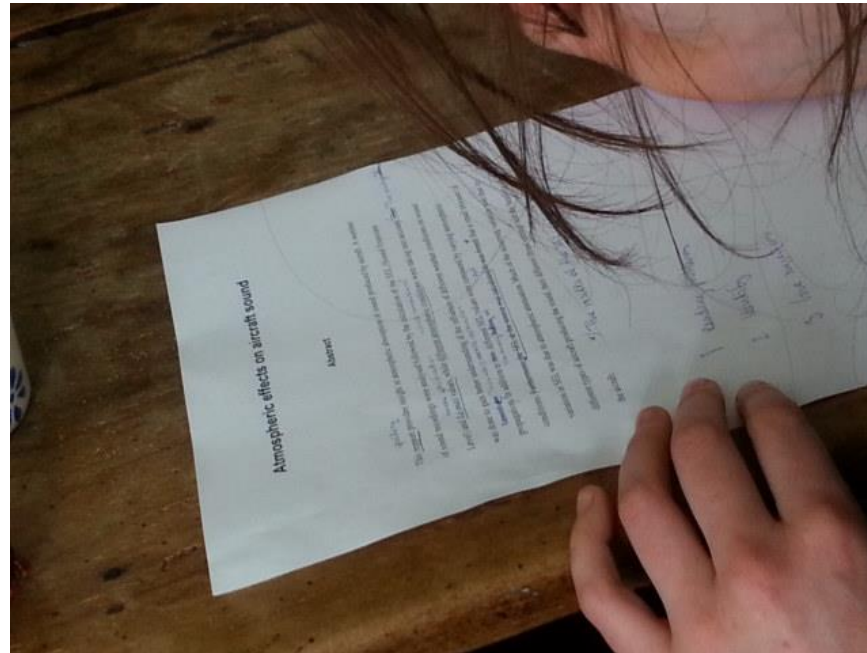
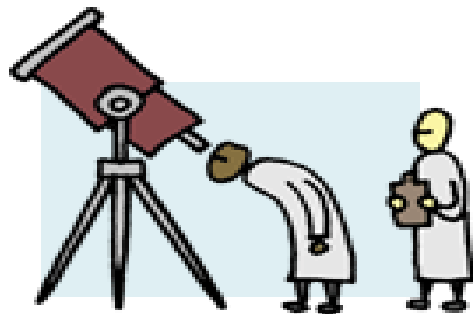


Tips for writing a useful peer review



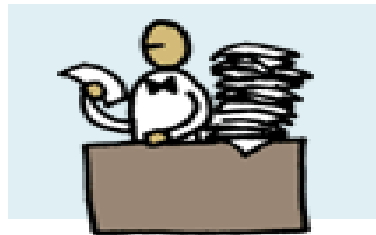
Scientific Writing in
AE2223-I Test, Analysis and Simulation
Centre for Languages and Academic Skills, TU Delft
May 2020



Scientists study something.



Scientists write about their results.



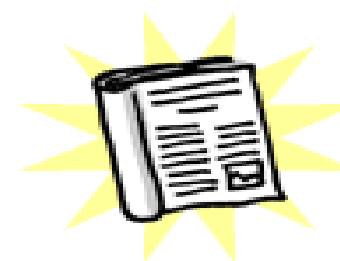
Journal editor receives an article and sends it out for peer review.



Peer reviewers read the article and provide feedback to the editor.



Editor may send reviewer comments to the scientists who may then revise and resubmit the article for further review. If an article does not maintain sufficiently high scientific standards, it may be rejected at this point.



If an article finally meets editorial and peer standards it is published in a journal.

The peer review process

This is you
(in this assignment)

Fill out the peer review form, use the checklist scientific articles

- Fill out the peer review form, based on your evaluation of the criteria for each category on the form
- The categories and criteria mentioned concisely on peer review form correspond to the checklist scientific articles
- Use the checklist for a more elaborate description of criteria

Peer review form

Elements
1. Abstract Readable as stand-alone text – Informative on research and results – Clear key words
Comments:
2. Introduction Background information on problem – Clear motivation for research – Research question stated clearly – Structure article discussed if necessary
Comments:
3. Method Sufficiently explained – Well argued for – Clear connection to research question/aim

Checklist scientific articles

Component
Elements
Abstract • can be read as a stand-alone text • is informative: includes conclusions and recommendations • key terms are clear to the intended readers
Introduction: • contains background information on the problem • indicates a motivation for the research / scientific gap / points out the problem in the present situation or previous research and argues why this issue should be resolved (including references) • contains a research question or hypothesis to show how this article contributes to solving the problem • indicates the structure of the article (unless AIMRaD is used)
Method • the method is sufficiently explained • the method used is well argued for • the method is shown to be connected to the research question / aim • if applicable: a 'Description of the data' section is used to describe the nature of the data used

Peer reviewing in four steps

1. Read the text
2. Identify reading problems
3. Explain the problem
4. Provide concrete suggestions

1. Read the text



- Read the entire article and form an impression of its coherence and main message
- Can you easily follow the storyline from research question to main conclusions? Do you think the main conclusions clearly correspond to the main question / issue raised in the introduction?
- Don't be tempted to start correcting lower order issues such as spelling and grammar straight away
- Then, provide comments for each category of the peer review form and spend most time on the categories you deem most relevant

2. Identify reading problems



In your comments, clearly indicate the place in the text where you encountered a problem:

“The second sentence of the second paragraph (“The effect of ... workload results”) lacks either commas or linking words, which makes the sentence difficult to read.”

3. Explain the problem

Clearly explain what the problem is, as is done in the following example:

“In section C, are the outliers ‘unavoidable due to imperfect experimental conditions’ or is there more at play, for example limitations in the PIV technology? I’d like to know a little more about that, so I can justify why you are replacing the outliers.”

4. Provide concrete suggestions

Indicate a possible solution to the problem:

“Both section C and D could be combined and called something like ‘Data processing’. Then the reader will first read about the physical setup, followed by an explanation of the method of data gathering, followed by the procedure to process all this data.”

Keep in mind: add useful comments only

“C Outlier detection and Replacement +
Data Smoothing and Vortex Identification:
No comments on style were needed here.”

(The comment above only adds more
words to the review – not a helpful addition)

Keep in mind: moderate your tone of voice



Which would you prefer?

1. “The actual description of SPIV is quite good! It’s a very detailed description! This I could reproduce myself now!”

or

2. “I would rephrase the 3rd sentence; it is too obvious and the words ‘lots of money’ do not seem very professional in my opinion. Give an approximation or leave it out.”

→ Try to strike a good balance between compliments and criticism; most writers will appreciate the constructive criticism in remark 2.

Be critical and respectful.

Keep in mind:

a reviewer is not an editor

- Of course, you may indicate spelling and language errors; however, you are not an editor
- Indicate a few examples of language and style issues you have come across in the category *Language and style* on the review form
- Leave it to the authors to thoroughly check the whole text
- Besides language, style and spelling, make sure you mostly focus on ‘higher order’ aspects such as cohesion and argumentation (is the main research question answered?), scientific gap, use of illustrations... etc.

Summary:

tips for writing a useful peer review

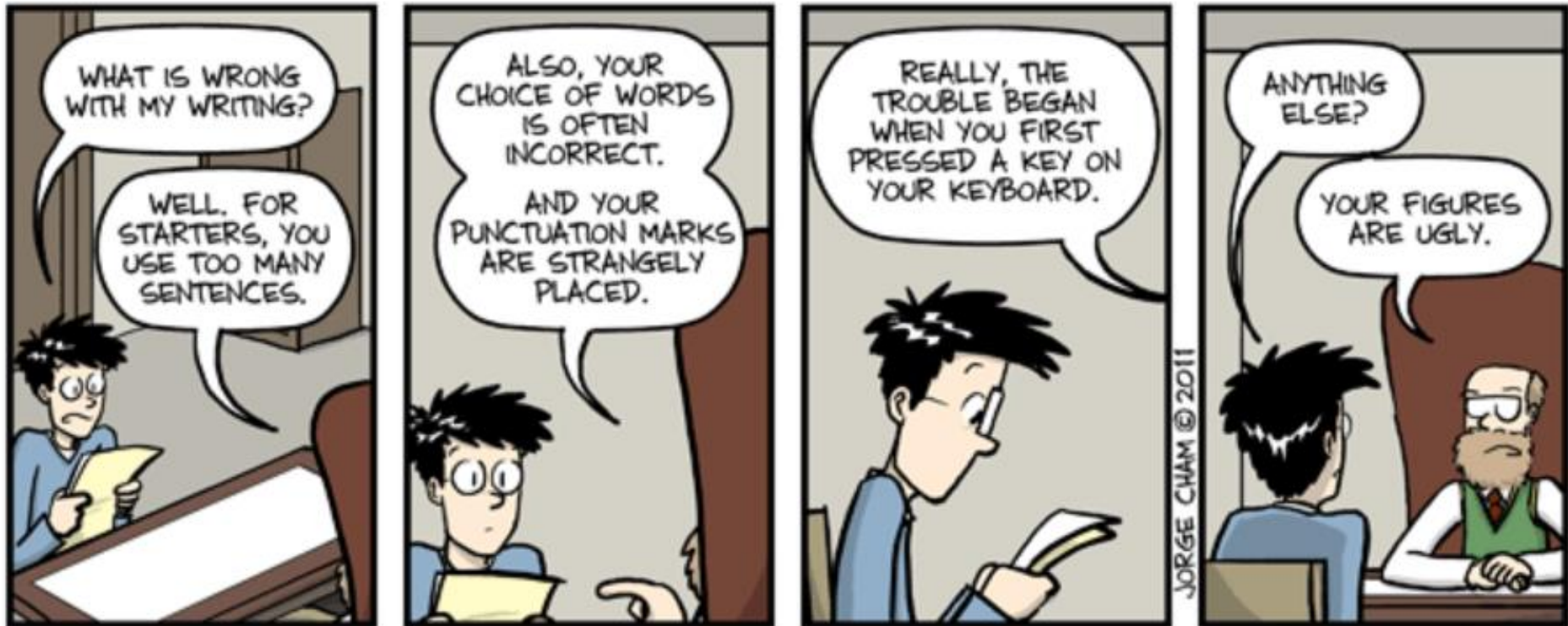
4 steps in a peer review:

1. Read the text
2. Identify reading problems
3. Explain the problem
4. Make concrete suggestions

Keep in mind:

- Add useful comments only
- Moderate your tone of voice (be critical *and* respectful)
- A reviewer is not an editor

Good luck!



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From:
"Piled Higher and Deeper" by Jorge Cham
www.phdcomics.com